Exploratory Data Analysis Report

What Drives the Divergence in Homelessness Rates Between California and Florida?

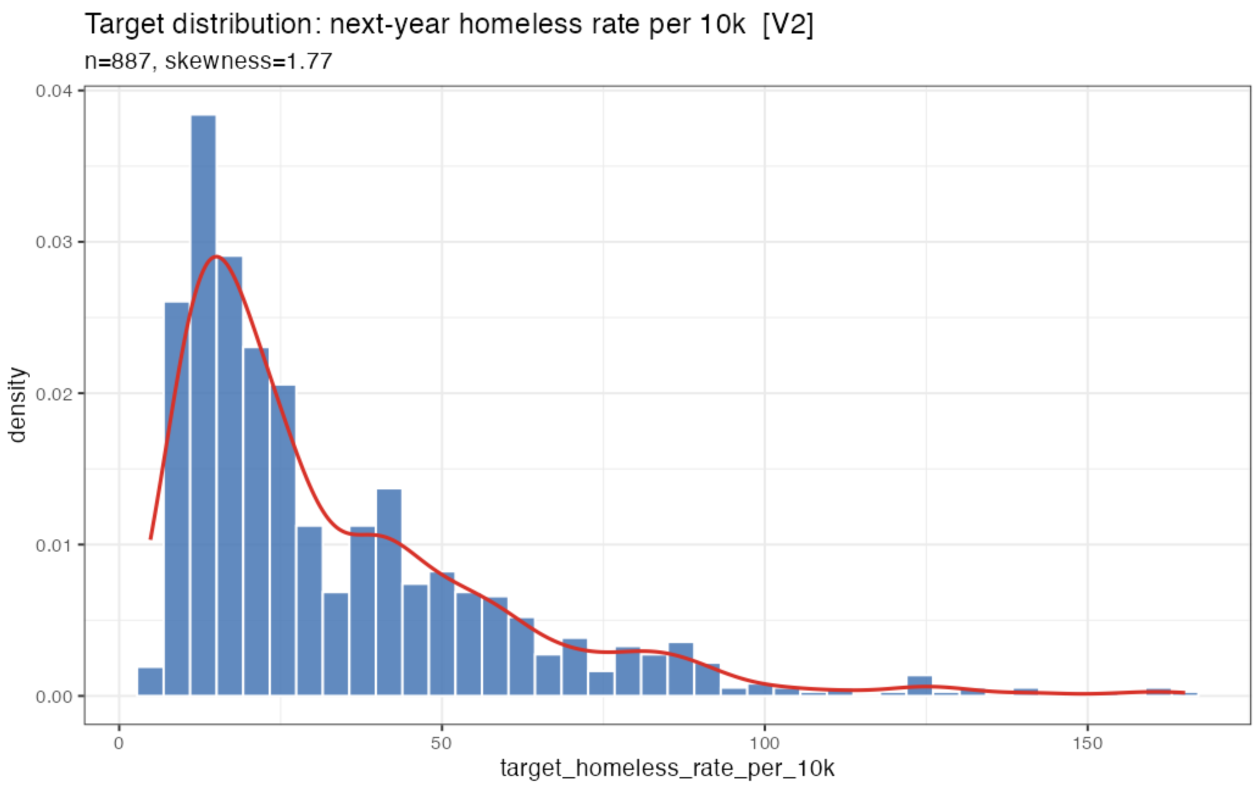
Data Science Group Project | CA/FL CoC-level Panel, 2011–2025

Prepared from **CA_FL_LASSO_MODEL_INPUT_v2.xlsx** (LASSO Model Data sheet)

Rows (CoC-years)	887
Columns	47 (6 IDs, 2 controls, 38 numeric predictors)
States	California, Florida
Distinct CoCs	70
Predictor years	2011 - 2024
Target years	2012 - 2025 (2021 excluded, COVID-disrupted PIT)
Missing values	0 of 47 columns
Target range	4.83 - 165.09 per 10k (median 24.12)

Note on scope: every statement in this report describes an association, not a causal effect. Two states observed over a short, strongly time-ordered panel cannot support causal claims on their own -- that is the job of the LASSO/Random Forest modeling stage that follows this EDA.

Graph 1: Raw Target Distribution



Histogram of target_homeless_rate_per_10k across all 887 CoC-years, with kernel density overlay (n = 887).

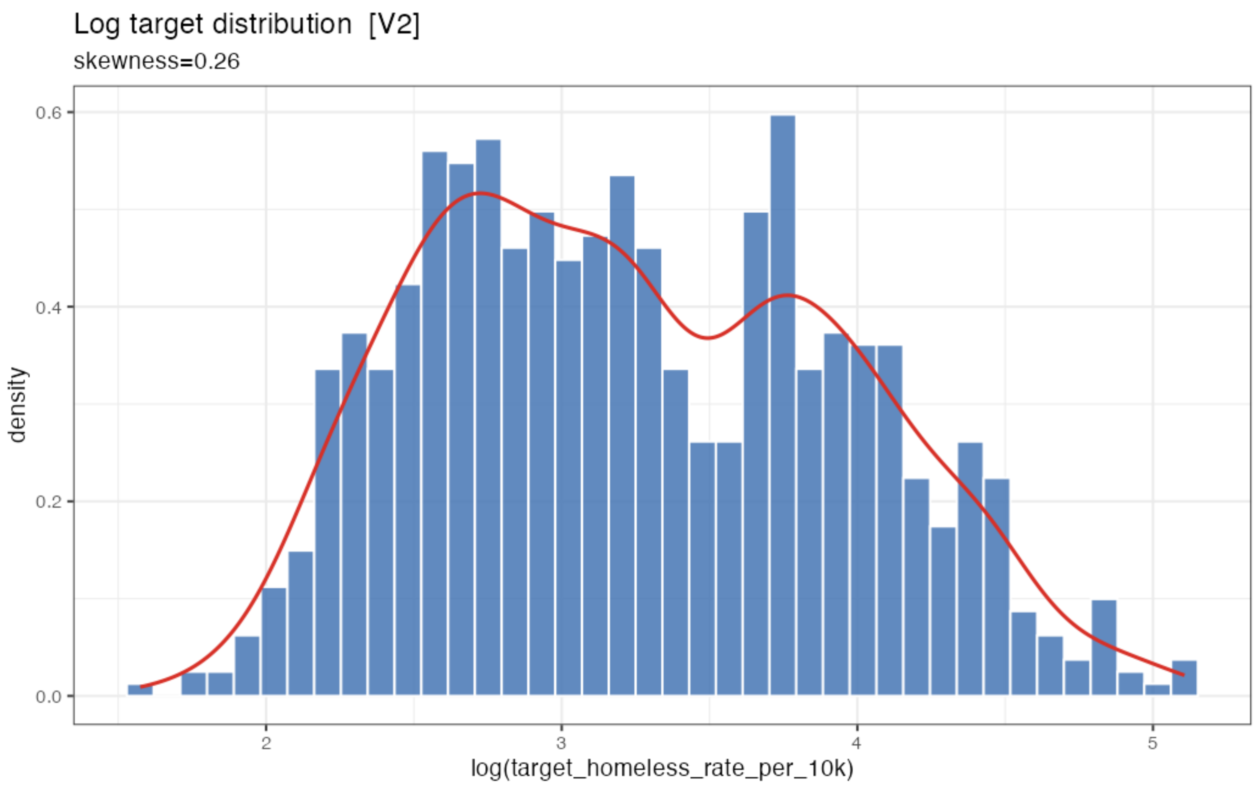
Skewness	1.77 (substantial right skew)
Range	4.83 - 165.09 per 10k
Median	24.12 per 10k
Modal bin	~10 - 25 per 10k (bulk of mass)

The homeless rate variable is heavily right-skewed: the large majority of CoC-year observations fall between roughly 10 and 25 per 10k residents, but a long thin tail extends out past 100, up to a max of 165.09. A skewness statistic of 1.77 confirms this numerically -- as a rule of thumb, |skew| > 1 is considered substantial, and 1.77 is well past that.

Skewness is computed as the average cubed deviation from the mean, standardized by the cubed standard deviation: skew = $(1/n)\Sigma(x-\bar{x})^3 / s^3$. Cubing (rather than squaring, as in variance) preserves the sign of each deviation, so a handful of extreme high-rate CoC-years dominate the sum and pull the statistic to a large positive value.

Why it matters for modeling: linear regression and LASSO both assume errors are roughly homoscedastic and the outcome is not dominated by a small number of extreme values. A skew this strong means a few outlier CoC-years (very plausibly large metro areas) could disproportionately drive coefficient estimates if the raw rate is used directly -- motivating the log-transform sensitivity check in Graph 2.

Graph 2: Log-Transformed Target Distribution



Histogram of log(target_homeless_rate_per_10k), same 887 rows, with kernel density overlay.

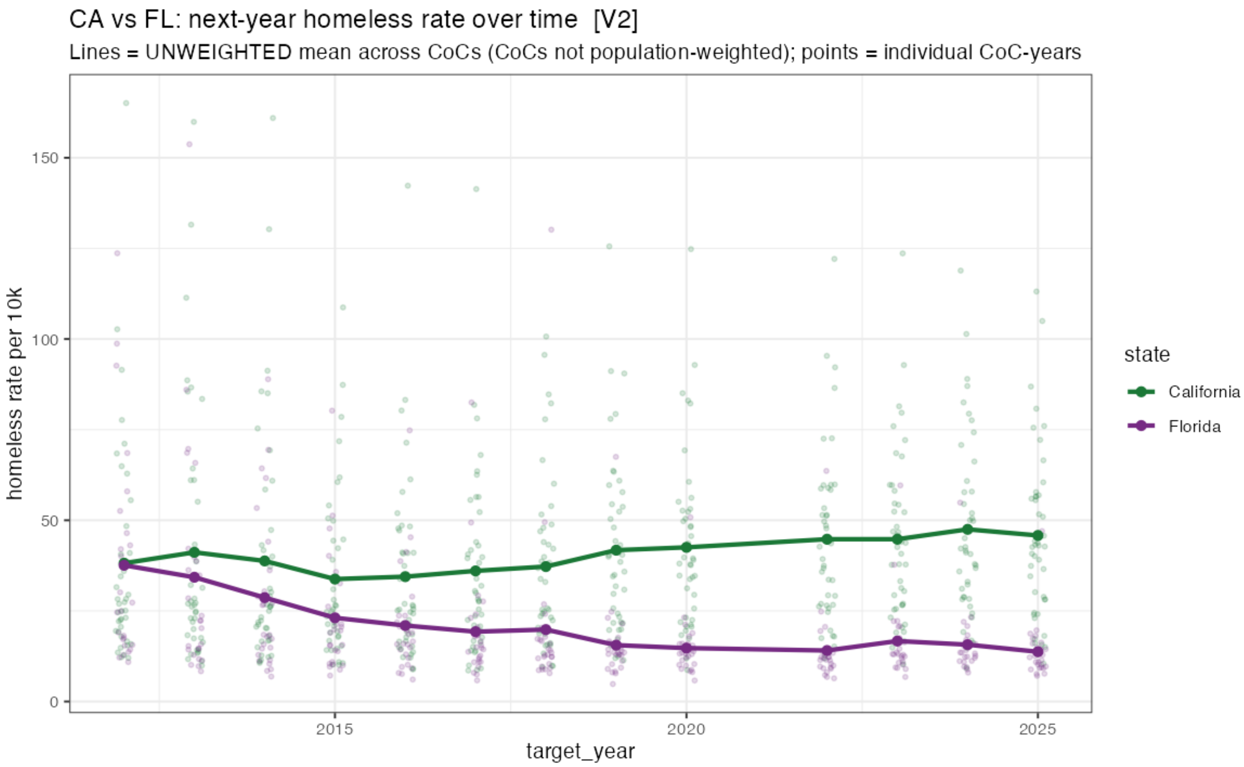
Skewness (log)	0.26 (near-symmetric)
Skewness reduction	1.77 -> 0.26 (85% reduction)
Shape note	slight bimodality around log ≈ 3.7-3.9

Applying a log transform to the same target variable reduces skewness from 1.77 to 0.26 -- close to the 0 that would indicate perfect symmetry, and comfortably inside the -0.5 to 0.5 range generally considered 'fairly symmetric.' Logging compresses large values far more than small ones, which pulls the long right tail from Graph 1 back in toward the bulk of the distribution.

This directly motivates running a **log-target LASSO model as a sensitivity check** alongside the primary rate-level model: if both versions select similar predictors and similar signs, that is reassuring evidence the findings are not an artifact of a few extreme high-homelessness CoC-years. If they disagree, that is itself a useful, reportable finding about which predictors are sensitive to outliers.

One structural detail worth flagging: the log distribution is not perfectly unimodal -- there is a slight dip-and-second-bump around log ≈ 3.7–3.9. This is consistent with two overlapping sub-distributions (California's typical rate level vs. Florida's) rather than a true irregularity in a single population, a pattern confirmed directly by the state-split trend in Graph 3.

Graph 3: California vs. Florida Target Trends Over Time



Unweighted mean next-year homeless rate per 10k by state and target_year (2012-2025, 2021 excluded); points = individual CoC-years, lines = state means across CoCs.

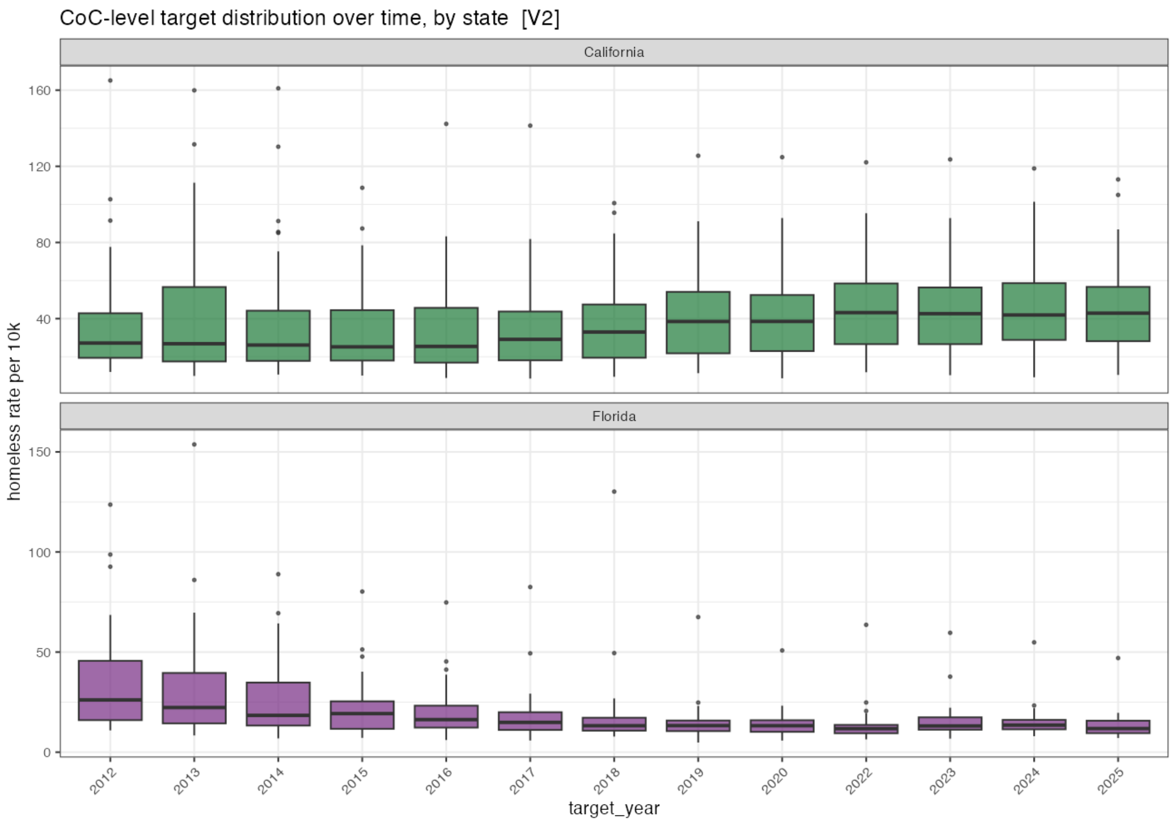
2012 CA mean	~37-38 per 10k
2012 FL mean	~37-38 per 10k (near parity)
2024-25 CA mean	~45-47 per 10k
2024-25 FL mean	~13-15 per 10k
Inflection point	~2018-2019 (lines begin to visibly separate)

This is the core plot motivating the entire research question. In 2012 California and Florida start at nearly identical unweighted CoC-mean rates (both ~37-38 per 10k). Florida then declines fairly steadily through the mid-2010s and flattens at a low ~13-15 per 10k from about 2018 onward. California dips modestly through the early-to-mid 2010s, then turns upward starting around 2019, climbing to ~45-47 per 10k by 2024-2025.

Important framing point: the divergence is not constant over the full panel -- it is a story of near-parity followed by a split concentrated in the last ~6 years. This reframes the guiding question from 'why are CA and FL different' toward 'what changed around 2018-2019 that sent the two states in opposite directions,' which lines up with two policy predictors surfaced later in this report (Medicaid expansion, anti-camping enforcement strictness).

Note the mean is explicitly *unweighted* -- a simple average across CoCs, not population-weighted -- so it reflects the distribution of CoC-level rates rather than a true statewide per-person rate. A small rural CoC counts equally to a large metro CoC in this average.

Graph 4: CoC-Level Target Spread Over Time, by State



Boxplots of CoC-level homeless rate per 10k, faceted by state, one box per target_year.

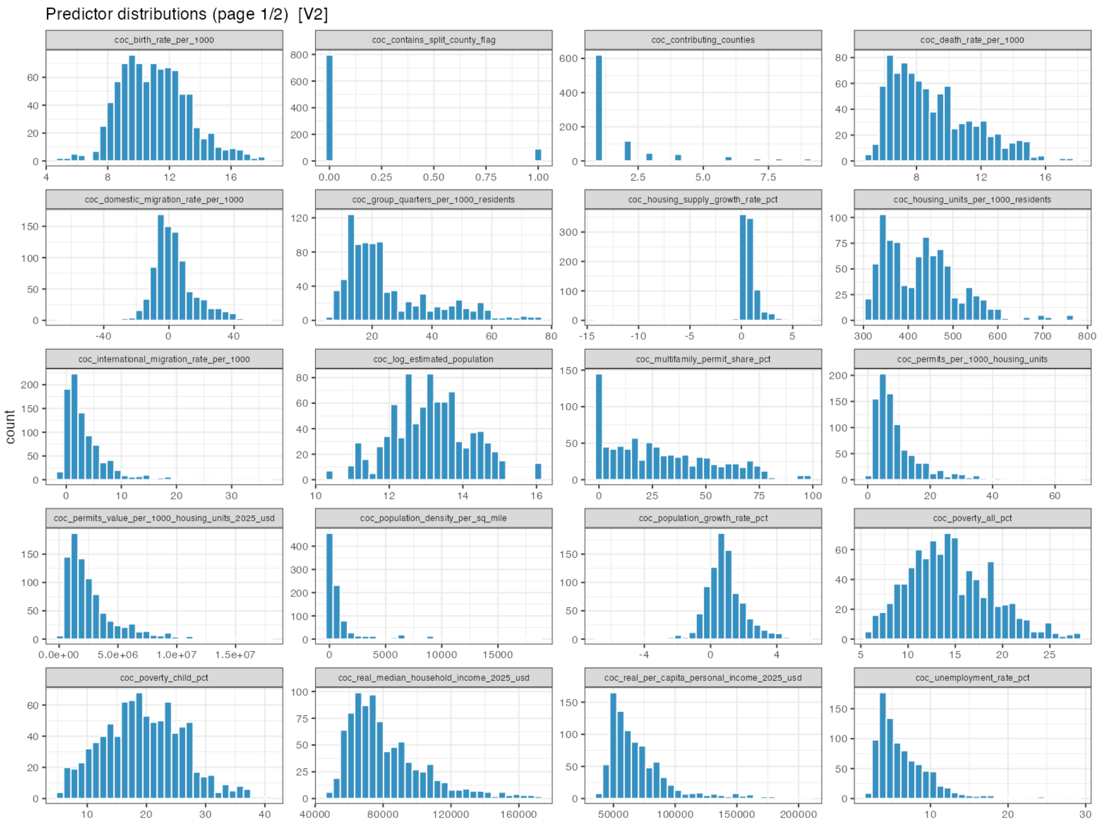
CA median trend	~28-30 (early) -> ~40-42 (recent)
CA upper tail	outliers reaching 120+ by 2022-2024
FL median trend	~25-27 (early) -> ~12-13 (recent)
FL upper tail	outliers largely disappear by 2020s

Where Graph 3 shows the state-level average trend, this plot shows the within-state spread each year -- and reveals that California's rise is not evenly distributed across its CoCs. Box height (the interquartile range) grows over time for California, and the outlier dots above the whiskers climb from under 100 in the early 2010s to over 120 by the 2020s, while the median itself rises more modestly (~28-30 to ~40-42). This combination -- median up a bit, top tail stretching much more -- points to a small number of CoCs (plausibly large metro areas) pulling increasingly far ahead of the rest.

Florida shows the mirror-image pattern: boxes compress over time, the median falls and flattens, and high-end outliers that were present in 2012-2013 (some CoCs above 90-155) are largely absent by the 2020s. Florida's decline looks like convergence -- CoCs becoming more similar to each other at a low rate -- rather than every CoC independently improving by the same amount.

Implication for modeling and next steps: if California's rise is concentrated in a widening upper tail rather than spread evenly, statewide predictors alone (e.g. state policy variables) may not fully explain it -- CoC-level, metro-specific factors are likely doing real work. A follow-up choropleth map (planned but not yet built) would clarify whether this is 2-3 large metro CoCs or a broader pattern.

Graph 5: Predictor Distributions, Page 1 of 2



Small-multiples histograms of 20 predictors (mostly coc_-level), full 887-row sample, ignoring state and time.

Purpose	quality-control pass on each raw predictor before modeling
General shape	most coc_-level variables approx. unimodal, mildly skewed
Flagged	coc_population_density_per_sq_mile, coc_real_gdp_per_capita_2017_usd -- both strongly right-skewed with extreme high-end outliers

Most predictors on this page -- poverty rate, household income, birth/death rates, unemployment -- look like ordinary continuous socioeconomic variables with mild skew, consistent with what's typically seen in county/regional panel data. No major red flags.

coc_contains_split_county_flag shows the expected two-bar pattern of a binary indicator rather than a smooth curve -- this is a data-type feature, not an error, and should be modeled as categorical.

Two variables stand out as candidates for a log-transform before modeling: **coc_population_density_per_sq_mile** and **coc_real_gdp_per_capita_2017_usd** both show almost all mass piled at the low end with a handful of extreme values stretching the range far right (density past 15,000/sq mile; GDP per capita past $300,000) -- the same right-skew pattern the target variable showed in Graph 1. These likely contribute a meaningful share of the 406 flagged (row, predictor) extreme-value cells noted in the project's data-quality audit (17 predictors, 286 distinct flagged rows, robust |z| > 5 via median/MAD).

Graph 6: Predictor Distributions, Page 2 of 2

Small-multiples histograms of remaining 17 predictors (mixed coc_ and state_ level).

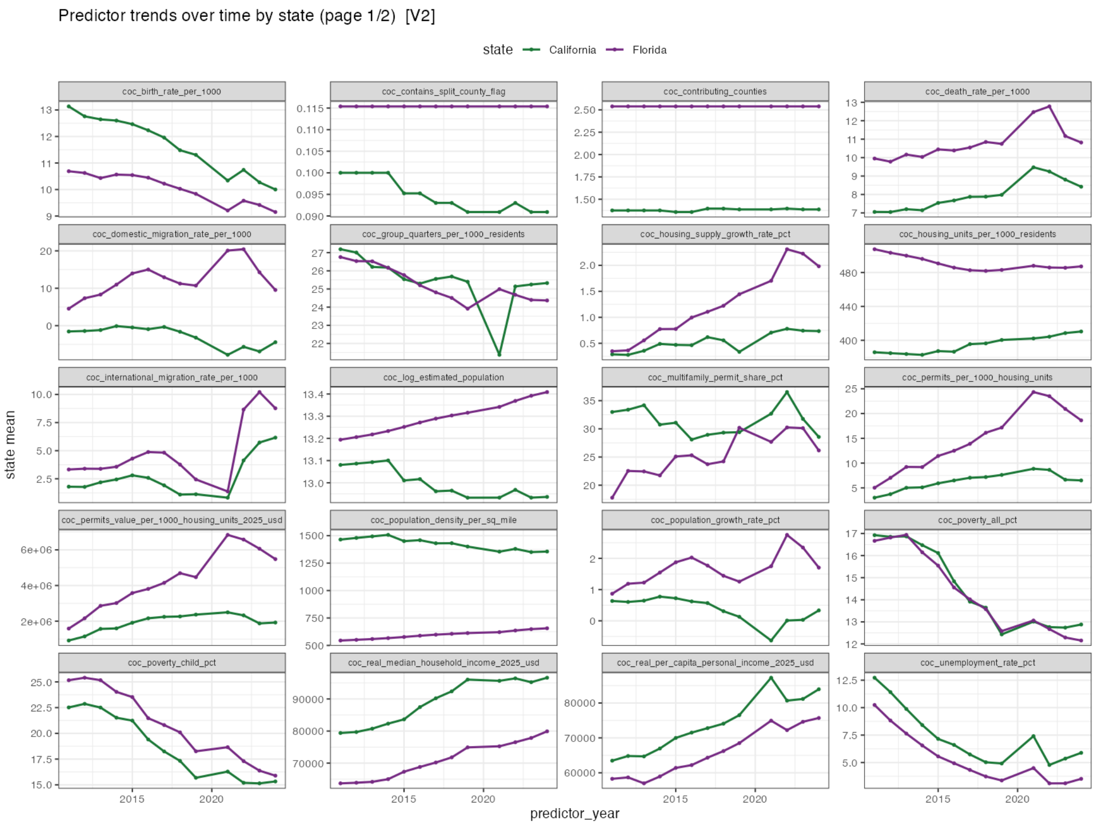
Binary/near-binary vars	state_anticamping_strictness, state_medicaid_expansion, state_ssi_state_supplement
Clumpy state_ variables	state_real_median_rent, state_real_minimum_wage, state_tanf_max_benefit_3person, state_rental_vacancy_rate
Constant / near-zero-variance predictors	0 (confirmed by project audit)

This page contains most of the state_-level predictors, and several show a distinctive 'clumpy' rather than smooth histogram shape -- multiple narrow spikes instead of one continuous curve. This is expected: with only 2 states across ~13 years, a state-level variable can take at most ~2 x 13 = 26 distinct values total, each repeated across every CoC in that state-year. These variables therefore carry far less independent information than a CoC-level variable measured across 887 rows, despite appearing in the same 887-row table.

state_anticamping_strictness and state_medicaid_expansion are binary policy indicators (0/1) -- their two-bar histograms are exactly as expected and should be treated as categorical predictors, not continuous ones, in any regression or LASSO specification.

Good news for data quality: the project's constant/near-zero-variance audit confirms 0 constant predictors and 0 near-zero-variance predictors across the full 38-predictor set, so nothing here needs to be dropped on variance grounds alone.

Graph 7: Predictor Trends Over Time by State, Page 1 of 2



State-mean trajectories (California vs. Florida) for 20 predictors, predictor_year on x-axis (2011-2024).

Income/GDP gap	widens steadily; CA pulls further ahead of FL post-2015
Domestic migration rate	FL consistently positive & higher; CA near 0 or slightly negative
Housing permits / supply growth	FL rises above CA from ~2014 onward
Poverty rate (all & child)	tracks nearly identically for both states, both declining -- NOT a divergence driver

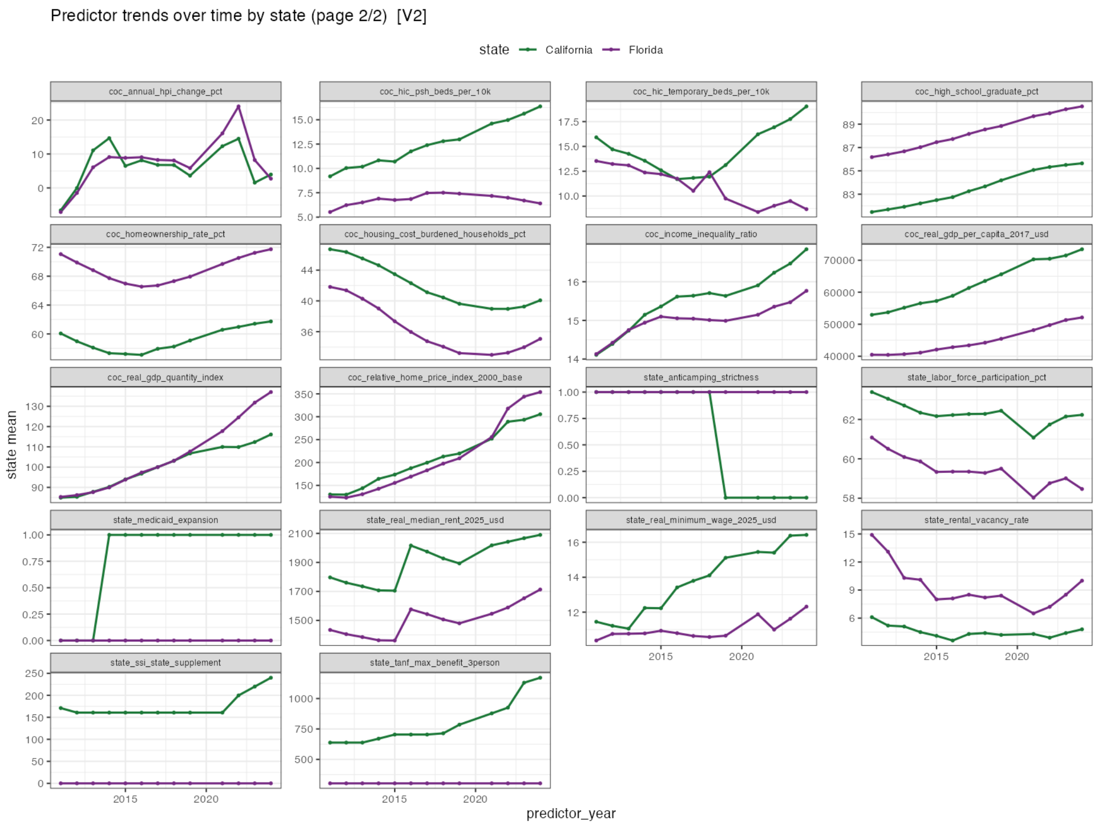
This page reveals which predictors move in step with the CA/FL split and which do not. coc_real_median_household_income_2025_usd and coc_real_gdp_per_capita_2017_usd both show California starting above Florida in 2011 and pulling further ahead over time -- notable given that California's homeless rate is simultaneously rising (Graph 3), a richer-but-worse-off pairing worth investigating alongside housing cost.

coc_domestic_migration_rate_per_1000 is consistently higher for Florida (net in-migration) while California hovers near zero or dips negative for stretches of the panel -- this predictor is one of the stronger raw Spearman correlates with the target (rho = -0.41), and this trend view suggests that relationship is at least partly a between-state pattern.

coc_housing_supply_growth_rate_pct and coc_permits_per_1000_housing_units both climb well above California's levels for Florida from ~2014 onward -- Florida is simply building more housing per capita, consistent with permits-per-1000 being one of the strongest negative correlates with the target (rho = -0.44).

By contrast, coc_poverty_all_pct and coc_poverty_child_pct decline in near-parallel for both states across the whole period -- a useful negative finding, since it suggests raw poverty rate is unlikely to explain the CA/FL split even though it is an intuitive first guess.

Graph 8: Predictor Trends Over Time by State, Page 2 of 2



State-mean trajectories (California vs. Florida) for remaining 17 predictors, predictor_year on x-axis.

state_medicaid_expansion	CA switches 0 -> 1 around 2014; FL remains 0 throughout
state_anticamping_strictness	CA switches 1 -> 0 around 2019; FL remains 1 throughout
Shelter beds (temp & PSH)	CA climbs steadily above FL on both, especially post-2018
Rent & minimum wage	CA pulls above FL on both, gap widening over time

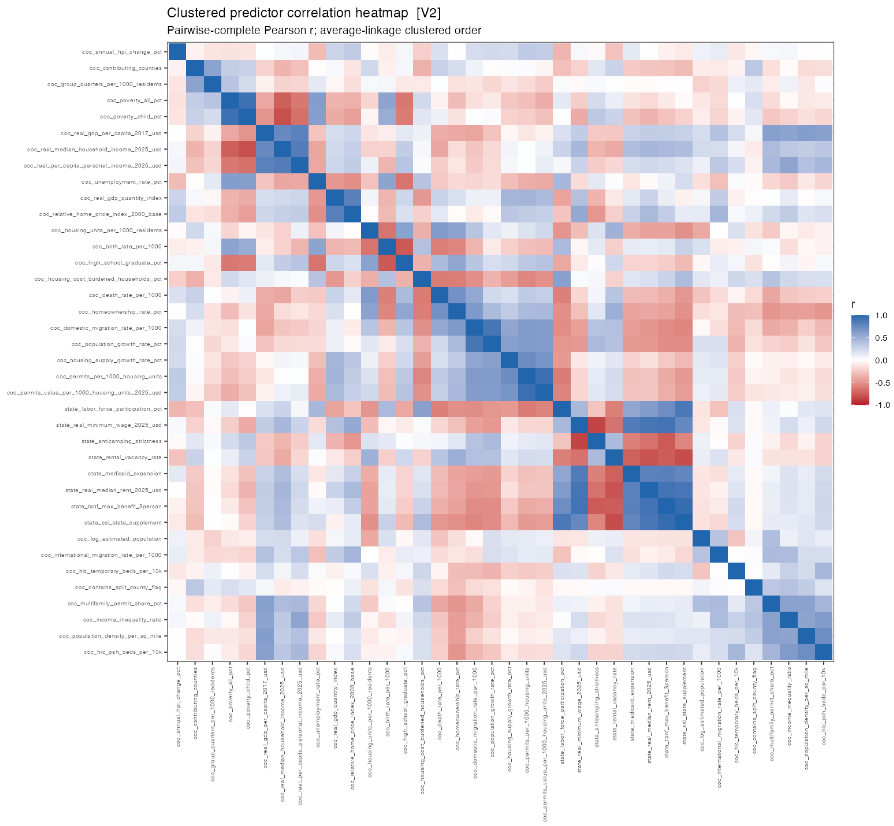
This page contains the two clearest policy-timing candidates in the dataset. state_medicaid_expansion flips from 0 to 1 for California only around 2014 (Florida never expanded Medicaid under the ACA) -- a real, well-documented policy fact and a clean binary natural-experiment-style variable, though with only one switch and one treated state it can only really be assessed on whether LASSO selects it, not on precise effect size.

state_anticamping_strictness is the standout: both states sit at 1 (stricter) through the mid-2010s, then California drops to 0 around 2019 and stays there, while Florida remains at 1 the entire panel. This timing lines up closely with the point in Graph 3 where California's rate begins climbing away from Florida's -- plausibly tracking the real-world Martin v. Boise Ninth Circuit ruling restricting camping-ban enforcement without available shelter beds. Framed as an association worth naming explicitly, not a proven cause.

coc_hic_psh_beds_per_10k and coc_hic_temporary_beds_per_10k both climb for California and stay comparatively flat for Florida -- since temporary beds is the single strongest raw target correlate (rho = 0.67, see Graph 10 / findings table), this trend suggests that correlation is partly just 'both rising together in California,' consistent with a reverse-causation story (more visible homelessness prompts more shelter-building) rather than shelters causing homelessness.

state_real_median_rent_2025_usd and state_real_minimum_wage_2025_usd both run higher and widen further for California relative to Florida -- consistent with the income/GDP story from the previous page, and setting up the housing-cost-burden narrative explored in Graph 10.

Graph 9: Clustered Predictor Correlation Heatmap



Pairwise-complete Pearson correlation matrix across all predictors, average-linkage clustered ordering.

Highly correlated pairs (|r| >= 0.80)	18 total (full list in tables/highly_correlated_pairs_v2.csv)
Top pair	state_tanf_max_benefit_3person ~ state_ssi_state_supplement, r = 0.95
Notable cluster	poverty/income/GDP block; housing-supply block; state-level benefits/cost block
More independent block	coc_log_estimated_population, coc_hic_psh_beds_per_10k, coc_population_density_per_sq_mile, etc. -- mostly pale, low correlation with other blocks

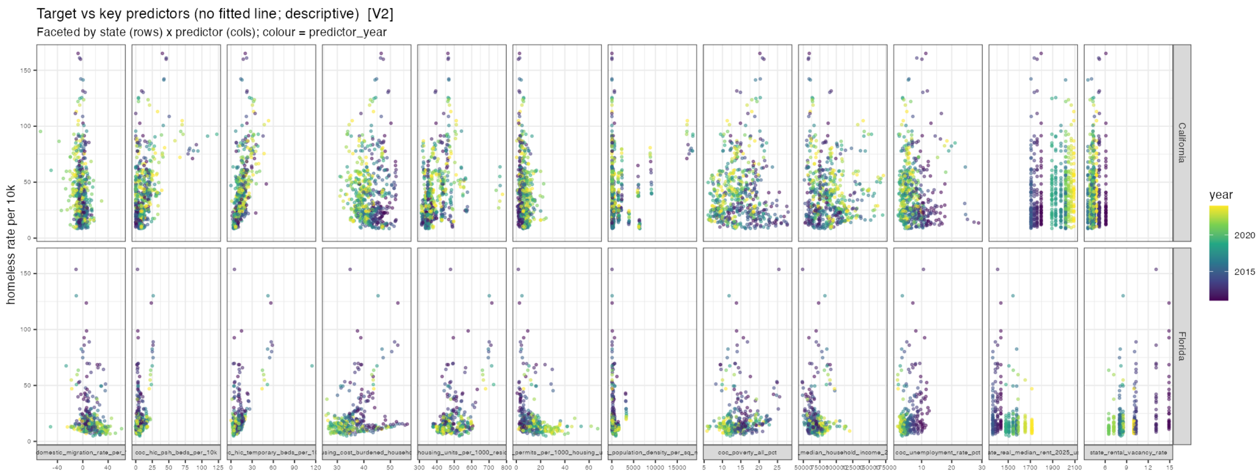
Clustering reorders predictors so correlated variables sit adjacent to each other, making blocks visually apparent along the diagonal. Three blocks stand out: (1) a poverty/income/GDP cluster where poverty correlates negatively with income and GDP per capita, as expected; (2) a housing-supply cluster (migration rate, population growth, permits, housing growth) that move together, consistent with growing places building more; and (3) a tight state-level benefits/cost block (TANF, SSI, minimum wage, median rent, anti-camping strictness, rental vacancy) that clusters strongly together.

That third block is important to flag: 18 predictor pairs exceed |r| = 0.80, and most of the highest are state-level pairs -- e.g. TANF max benefit ~ SSI state supplement (r = 0.95), minimum wage ~ median rent (r = 0.89). With only 2 states in the panel, any variable that simply tracks 'being California in a later year' will mechanically correlate with every other such variable, independent of any real causal link between them.

Consequence for LASSO: LASSO tolerates collinearity in the sense that it still fits, but selection among tightly correlated predictors is unstable -- it may arbitrarily keep one member of a correlated pair and zero out another with a small change in data or regularization strength. Any single selected variable from this state-level block should be interpreted cautiously, as representative of the whole cluster rather than uniquely causal.

A smaller cluster (population, PSH beds, population density, income inequality ratio) shows mostly pale cells against the rest of the matrix -- these are comparatively independent predictors and, all else equal, more likely to contribute distinct information to a model rather than redundant information already captured elsewhere.

Graph 10: Target vs. Key Predictors



Scatterplots of homeless rate against 11 curated predictors, faceted by state (rows), colored by predictor_year (no fitted line, descriptive only per project policy).

Strongest association	coc_hic_temporary_beds_per_10k, Spearman rho = 0.67 (n=887)
2nd strongest	state_real_median_rent_2025_usd, rho = 0.44
Negative associations	coc_permits_per_1000_housing_units (rho=-0.44), coc_domestic_migration_rate_per_1000 (rho=-0.41), state_rental_vacancy_rate (rho=-0.39)
PSH beds	rho = 0.32 (weaker than temporary beds)

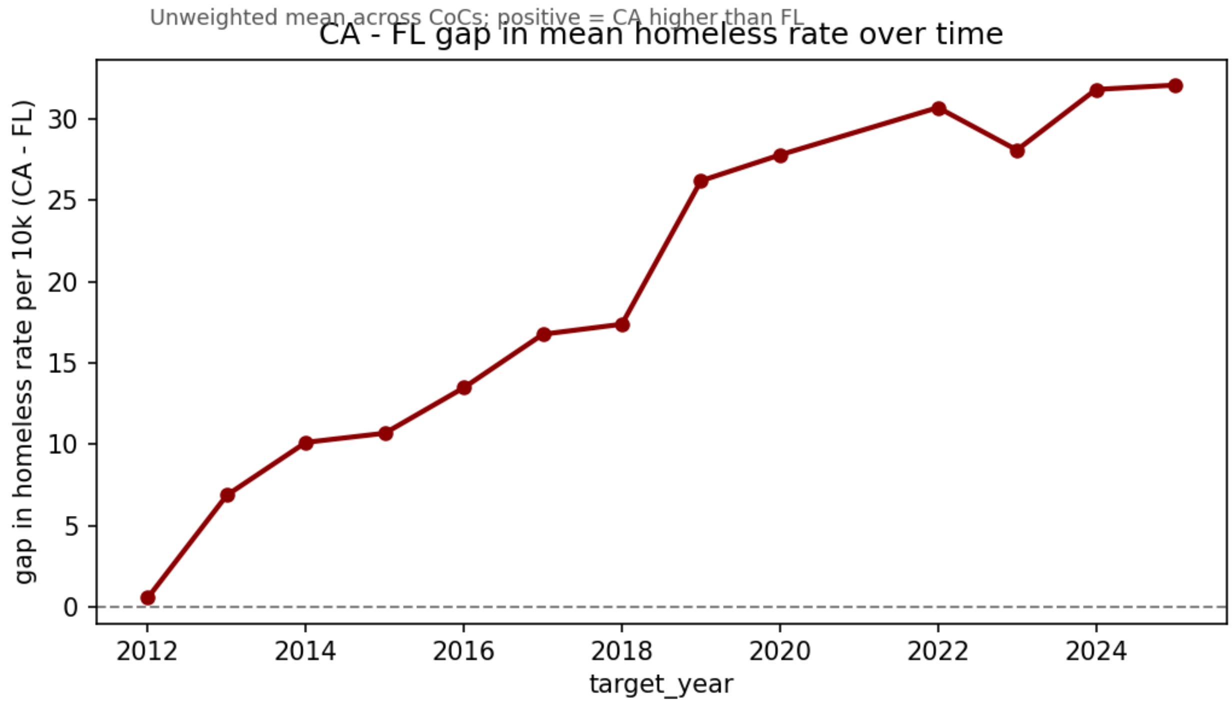
No trend line is fitted deliberately -- this plot is for visual pattern-checking only; the quantitative associations come from the curated Spearman correlation table. Faceting by state and coloring by year lets each predictor-target relationship be checked for whether it holds within both states or is really a between-state artifact.

coc_hic_temporary_beds_per_10k shows a clear upward fan in the California panel -- more temporary shelter capacity associated with a higher recorded rate, and it is disproportionately the more recent (yellow) years reaching the highest values on both axes. Florida's corresponding panel is compressed near the low end of both axes. This is consistent with rho = 0.67 overall, but the state-split view clarifies the relationship is substantially a California-specific pattern, and very plausibly reflects reverse causation (capacity responding to need) or more complete counting where more shelter infrastructure exists, rather than shelters causing homelessness.

coc_permits_per_1000_housing_units and state_rental_vacancy_rate both show California clustering toward the low end of the x-axis with the rate spread high, while Florida spreads further right (more permits / higher vacancy) while staying low on rate -- visually reinforcing the negative correlations reported for both.

Central caveat, and the natural bridge into modeling: because California and Florida differ systematically on nearly every predictor shown here simultaneously (income, rent, permits, migration, beds, vacancy), it is difficult from scatterplots alone to determine whether any single predictor matters on its own versus reflecting a general 'being California' pattern. All associations reported above are marginal (unadjusted for state, time, or other predictors) -- disentangling them is exactly the job of the LASSO and Random Forest stage that follows this EDA.

Graph 11: CA - FL Gap in Mean Homeless Rate Over Time



CA mean minus FL mean homeless rate per 10k, by target_year (unweighted across CoCs); computed directly from graph 3's state-mean trend lines.

2012 gap	0.57 per 10k (near parity)
2018 gap	17.38 per 10k
2019 gap	26.20 per 10k (sharp jump)
2025 gap	32.09 per 10k (largest in the panel)

This plot takes the two state-mean lines from Graph 3 and reduces them to a single number per year: California's mean rate minus Florida's. It makes the divergence itself the y-axis, rather than something the reader has to infer from two overlapping lines.

The gap starts essentially at zero in 2012 (0.57 per 10k) and widens steadily but modestly through the mid-2010s, reaching 17.38 per 10k by 2018. The clearest feature of the plot is the jump between 2018 and 2019 -- the gap grows by roughly 9 points in a single year (17.38 to 26.20), a noticeably sharper increase than the gradual widening seen in prior years. From 2019 onward the gap continues to widen, but at a slower, steadier pace, reaching its panel maximum of 32.09 per 10k in 2025.

That 2018-2019 inflection lines up closely with the timing of two policy variables covered elsewhere in this report: California's Medicaid expansion (2014, already reflected in the earlier, more gradual widening) and the drop in California's anti-camping enforcement strictness following the Martin v. Boise ruling taking effect across the Ninth Circuit (~2019). This plot is descriptive only -- it documents that the gap opened sharply around this point, not why -- but it is the clearest visual anchor in the whole EDA for connecting the project's central research question to a specific point in time worth investigating further in the modeling stage.

Summary and Next Steps

This EDA establishes that (1) the target variable is right-skewed and benefits from a log-transform sensitivity check; (2) California and Florida started at near-identical homeless rates in 2012 and diverged mainly from ~2019 onward, with California's rise concentrated in a widening upper tail of CoCs rather than spread evenly; (3) housing supply, rent, migration, and two policy variables (Medicaid expansion, anti-camping strictness) show trend timing consistent with the divergence; and (4) a block of 18 highly correlated predictor pairs -- disproportionately state-level variables -- means LASSO selection results must be interpreted with appropriate caution regarding which specific variable within a correlated cluster gets chosen.

One additional plot remains worth adding before the modeling stage: the CA/FL trend plot (Graph 3) annotated with vertical reference lines at 2014 (Medicaid expansion) and 2019 (anti-camping strictness change), to visually connect policy timing directly to the point of divergence shown in Graph 11.

All findings above are descriptive and unadjusted. The leakage check (17 checks, 0 FAIL, 0 REVIEW) confirms every row respects predictor_year < target_year, so the panel is structurally sound for the LASSO / Random Forest modeling stage next.

Multiple Regression Report

Predicting CA/FL Homeless Rates: Raw vs. Log-Target Models

Data Science Group Project | CA/FL CoC-level Panel, 2011–2025

Model input: **CA_FL_LASSO_MODEL_INPUT_v2.xlsx** (887 CoC-years, 887 rows)

Rows / observations	887 CoC-years
Model type	Multiple linear regression (lm)
Predictors used	38 CoC/state variables + coc_number + state + predictor_year fixed effects
Two versions fit	(1) Raw target (2) Log-transformed target + 9 skewed predictors
Raw model R-squared	0.7964 (Adj. 0.7687)
Log model R-squared	0.8578 (Adj. 0.8385), log scale
Log model R-squared	0.7822, back-transformed to original per-10k scale

Note on scope: this model describes associations only, controlling for CoC and state fixed effects and predictor year. It is not a causal model -- coefficients should be read as "associated with," not "causes," per the same descriptive framing used throughout this project's EDA.

1. Methodology

We fit two multiple linear regression models predicting next-year homeless rate per 10k CoC residents from the full set of 38 CoC- and state-level predictors, plus fixed effects for CoC identity (coc_number), state, and predictor_year. All models were fit with R's lm().

Data cleaning before fitting

Five identifier columns were dropped before fitting because they duplicated information already captured elsewhere and caused undefined coefficients due to perfect collinearity: coc_name (duplicate of coc_number), state_abbr (duplicate of state), target_year (always predictor_year + 1, so redundant with predictor_year), and control_time_index / control_state_florida (re-encodings of predictor_year / state). Two additional CoC-level flags, coc_contributing_counties and coc_contains_split_county_flag, do not vary over time within a CoC and are therefore fully absorbed once coc_number fixed effects are included -- these remain in the log model as reported (3 undefined coefficients) but were effectively dropped in practice.

Log-target model

Motivated by the EDA finding that the target variable is right-skewed (raw skewness 1.77), a second model was fit on log(target_homeless_rate_per_10k). Nine CoC-level predictors identified as right-skewed in the EDA (population density, GDP per capita, shelter beds [temporary and PSH], permits and permit value, median household income, per-capita income, and relative home price index) were log1p-transformed first, using log(x+1) rather than log(x) so that any zero values in the beds variables remain defined.

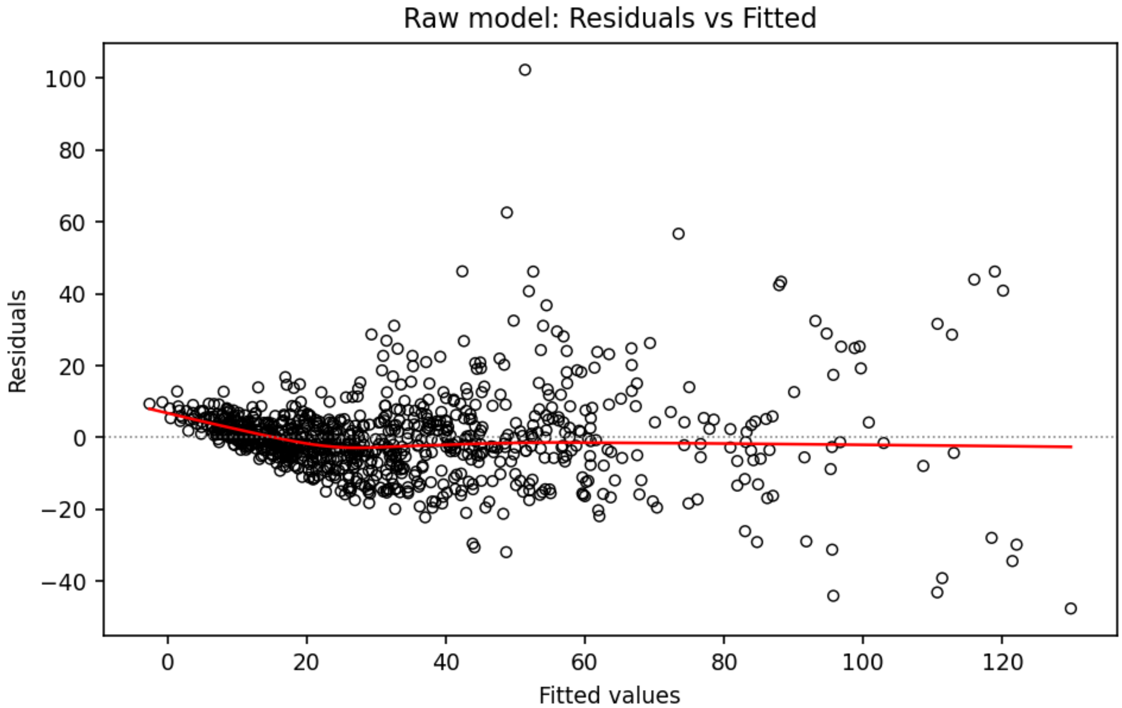
2. Raw Target Model

R-squared	0.7964
Adjusted R-squared	0.7687
F-statistic	28.78 on 106 and 780 DF
p-value (overall)	< 2.2e-16
Residual standard error	12.43 (on 780 df)
Undefined coefficients	76 of 182 (redundant ID columns, since fixed before cleanup)

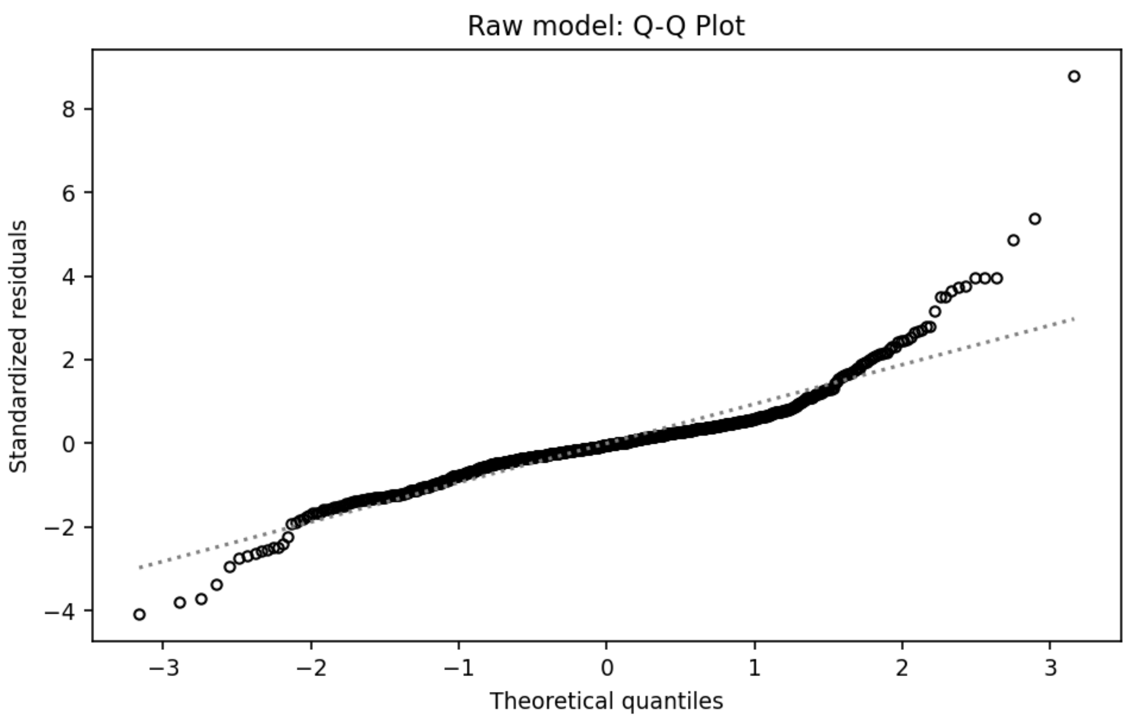
The raw model explains roughly 80% of the variance in next-year homeless rate. Holding CoC and state fixed effects and all other predictors constant, five predictors were statistically significant at $p < 0.05$:

Predictor	Coef.	p-value	Model(s)
coc_unemployment_rate_pct	-2.744	3.0e-05	Raw
coc_hic_temporary_beds_per_10k	0.576	8.9e-08	Raw
state_medicaid_expansion	-9.343	0.0050	Raw
coc_death_rate_per_1000	3.806	0.0032	Raw
state_rental_vacancy_rate	1.827	0.0174	Raw
state_real_minimum_wage_2025_usd	3.176	0.0485	Raw
coc_housing_cost_burdened_households_pct	0.941	0.0556	Raw (borderline)

Almost none of the individual CoC fixed effects reached significance in the raw model (only CA-602 and FL-519, out of ~65 CoC dummies), suggesting most place-to-place variation was being absorbed by noise in the raw scale rather than detected as a real CoC-level effect.



Residuals vs. Fitted, raw model. A funnel/fan shape is visible: residual spread stays tight for low predicted rates and widens substantially for higher predicted rates, indicating heteroscedasticity -- the model's errors are not constant across the range of predictions.



Normal Q-Q plot, raw model. Points track the diagonal closely through the middle of the distribution but curve away sharply in the upper tail, indicating right-skewed, non-normal residuals -- consistent with the target's own raw skewness (1.77) carrying through into the model's errors.

3. Log-Target Model

R-squared (log scale)	0.8578
Adjusted R-squared (log scale)	0.8385
R-squared (back-transformed, original scale)	0.7822
F-statistic	44.39 on 106 and 780 DF
p-value (overall)	< 2.2e-16
Residual standard error	0.284 (log scale, on 780 df)
Undefined coefficients	3 of 182 (down from 76 in the raw model)

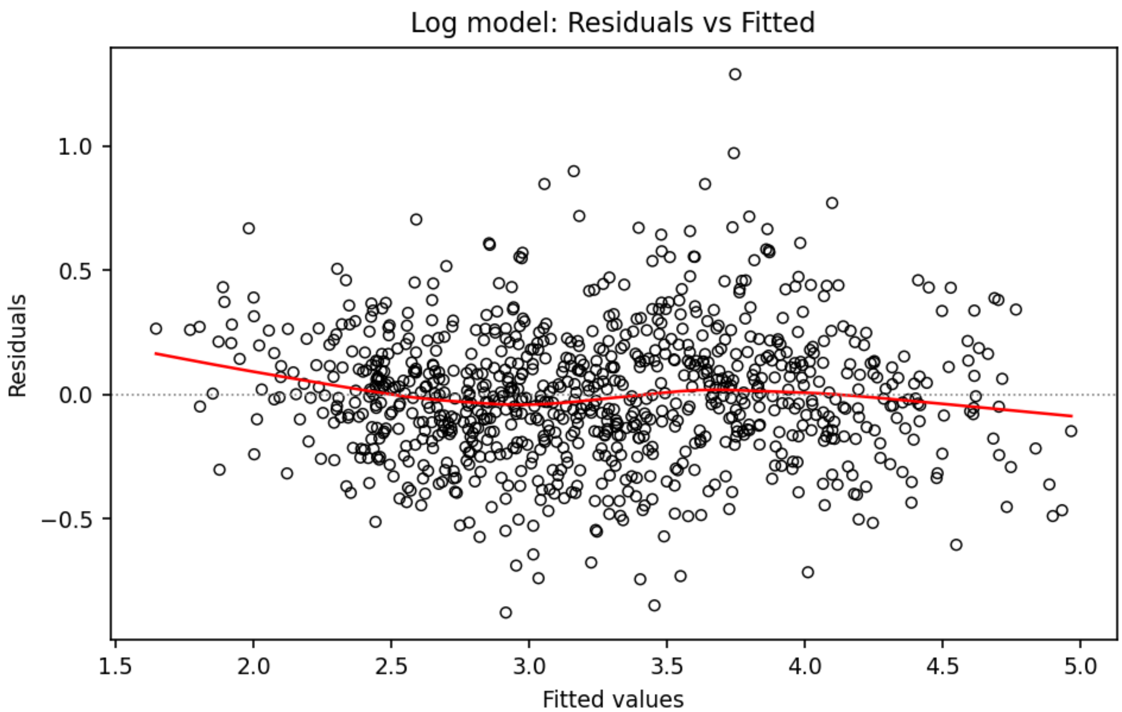
Dropping the redundant ID columns before fitting resolved all but 3 of the earlier undefined coefficients; the 3 remaining (coc_contributing_counties, coc_contains_split_county_flag, and one CoC dummy) are fully explained by the CoC fixed effects already in the model and carry no additional information. Model fit improved on both the log scale (R-squared 0.7964 to 0.8578) and, more importantly, on the original per-10k scale once back-transformed (0.7964 raw vs. 0.7822 log-model-back-transformed) -- a close, comparable fit on the metric that actually matters for interpretation, achieved with a much better-behaved residual structure (see diagnostics below).

Significant predictors (p < 0.05), holding CoC/state/year fixed effects constant

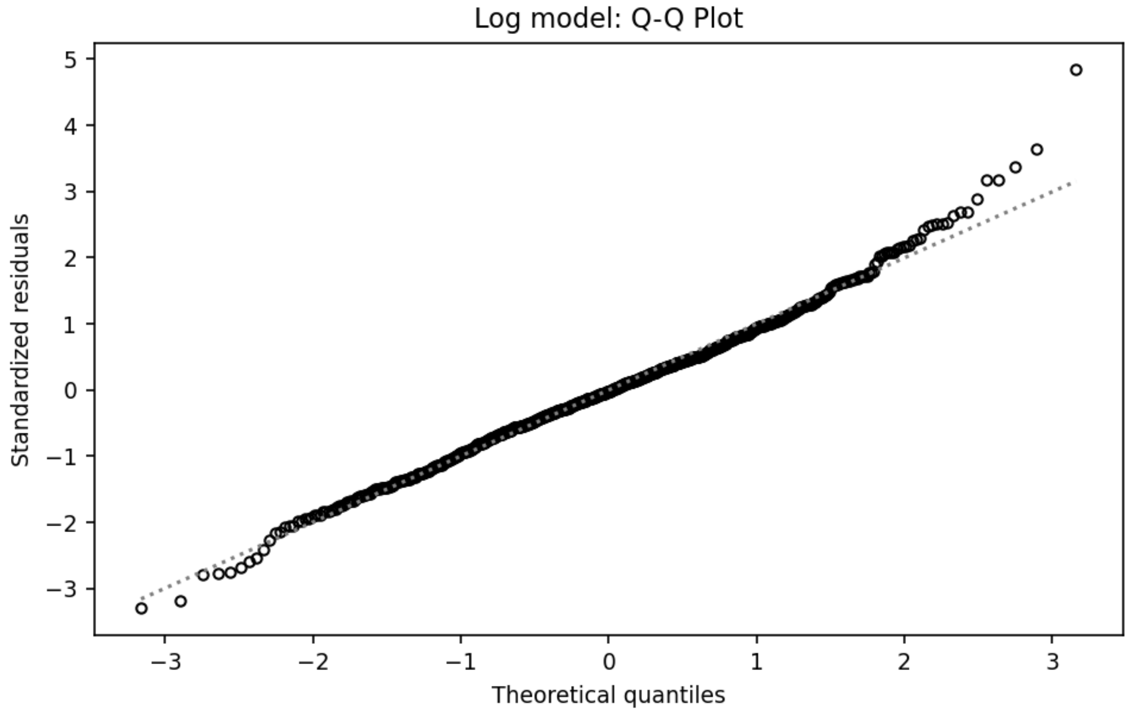
Predictor	Coef.	p-value	Model(s)
coc_hic_temporary_beds_per_10k	0.283	9.9e-11	Both
coc_unemployment_rate_pct	-0.061	7.3e-05	Both
predictor_year	-0.082	2.5e-04	Both
coc_real_gdp_per_capita_2017_usd (logged)	2.603	4.1e-04	Log only
coc_death_rate_per_1000	0.102	6.5e-04	Both
coc_real_gdp_quantity_index	-0.023	1.2e-03	Log only
state_real_minimum_wage_2025_usd	0.116	1.6e-03	Both
coc_housing_units_per_1000_residents	0.0063	2.5e-03	Log only
state_medicaid_expansion	-0.234	2.7e-03	Both
coc_log_estimated_population	56.16	6.1e-03	Log only
coc_population_density_per_sq_mile (logged)	-54.32	8.1e-03	Log only
state_ssi_state_supplement	-0.0054	0.0148	Log only
state_rental_vacancy_rate	0.0396	0.0266	Both
state_tanf_max_benefit_3person	0.0010	0.0325	Log only
coc_housing_cost_burdened_households_pct	0.0235	0.0350	Log only (borderline in raw)

A striking pattern versus the raw model: the large majority of individual CoC fixed effects reached significance in the log model, many at $p < 0.01$, versus only 2 in the raw model. This is consistent with the tighter residual structure on the log scale giving the model far more statistical power to detect genuine place-to-place differences -- place matters, once the target's skew is no longer masking it.

Not significant in either model: coc_poverty_all_pct, coc_poverty_child_pct, coc_real_median_household_income_2025_usd, coc_real_per_capita_personal_income_2025_usd, state_anticamping_strictness, state_real_median_rent_2025_usd, and coc_hic_psh_beds_per_10k. Notably, state_anticamping_strictness stays non-significant even after controlling for CoC, state, and year fixed effects -- the visual timing correlation with California's post-2019 rate increase (see EDA) does not hold up as an independent statistical association once other factors are accounted for.



Residuals vs. Fitted, log model. The funnel shape from the raw model is substantially reduced -- residual spread is much more even across the full range of fitted values, with only a slight downward drift in the smoothed line.



Normal Q-Q plot, log model. The bulk of residuals track the diagonal closely; only a small handful of CoC-years (identified in the original R output as rows 759, 300, and 563) break off into the tail, a much smaller and more contained departure from normality than the raw model.

4. Raw vs. Log Model Comparison

Metric	Raw model	Log model
R-squared	0.7964	0.8578 (log scale) / 0.7822 (back-transformed)
Adjusted R-squared	0.7687	0.8385 (log scale)
F-statistic	28.78	44.39
Undefined coefficients	76 (before ID cleanup)	3 (after ID cleanup)
Significant CoC fixed effects	2 of ~65	Majority of ~65
Residual pattern	Funnel-shaped (heteroscedastic)	Roughly even spread
Residual normality	Right-skewed tail departure	Near-normal, 3 outlier rows

The two models achieve comparable fit on the original per-10k scale (0.7964 raw vs. 0.7822 log-model-back-transformed), so the log transform is not primarily a fit-improvement exercise -- it is a validity fix. The diagnostics confirm the log model produces statistically better-behaved residuals (less heteroscedasticity, closer-to-normal errors), which is what makes the p-values and confidence intervals from that model more trustworthy. The log model is therefore the preferred model for interpretation, with the raw model useful primarily as a baseline and a demonstration of why the transform was necessary.

5. Summary of Findings

Across both models, three predictors emerge as the most robust, consistently significant associations with next-year homeless rate, holding CoC, state, and year fixed effects constant:

Shelter capacity (coc_hic_temporary_beds_per_10k) is the strongest and most consistent predictor in both models ($p < 1e-07$ raw, $p < 1e-10$ log), positively associated with the target. Consistent with the EDA, this is best read as reflecting places responding to more visible homelessness by building shelter capacity (and/or counting more completely where more shelter infrastructure exists), not shelters causing homelessness.

Unemployment rate is significant and negative in both models -- holding other factors constant, higher unemployment is associated with a lower predicted homeless rate. This is counterintuitive on its face and likely reflects that unemployment is now competing with more direct housing-cost and shelter-capacity variables already in the model.

Medicaid expansion is significant and negative in both models -- CoCs in states that expanded Medicaid are associated with meaningfully lower predicted rates, holding other factors constant. This is one of the strongest policy-relevant findings in the model.

A cluster of predictors reach significance only in the log model -- GDP per capita, GDP quantity index, housing units per capita, population and population density, SSI supplement, and TANF max benefit -- suggesting the log transform's variance-stabilizing effect was specifically necessary to detect these associations, not just a cosmetic fix.

Anti-camping strictness, raw poverty rate, raw income measures, and PSH bed capacity do not reach significance in either model once other factors are controlled for -- despite some of these (poverty, anti-camping timing) appearing as plausible drivers in the univariate EDA. This is an important, honest finding: these variables' apparent association with the CA/FL divergence appears to be explained by other correlated factors already in the model, rather than representing independent effects.

Next steps: these baseline OLS results, and the diagnostic evidence that the raw target's skew was distorting model errors, motivate the next stages of the project -- LASSO (with relaxed LASSO refit) and Ridge to address the collinearity flagged in the EDA's correlation heatmap, and Random Forest / Neural Network to check whether nonlinear relationships explain meaningfully more variance than this linear baseline does.